

Microsoft Azure: The platform offered by Microsoft Azure offers multiple options for configuring a cloud-based data management system. Additionally, it offers practical analytical tools for usage with data saved on Azure. Azure offers a robust toolkit for managing various database and data warehouse designs.

- Offers several storage services, including Blob, Table, Queue and File
- Private cloud deployments are available
- NoSQL-style table storage options
- Azure Data Explorer (ADX) enables a user to perform real-time analysis of very large streaming data without the need for preprocessing

Google Cloud Platform (GCP):

GCP offers a large set of tools for cloud-based data management. It comes with a workflow manager that ties the different components together. It offers:

- Google Data Studio for graphical user interface-based analysis and dashboard construction
- Cloud Datalab for code-based data science
- ML engine for advanced analysis through ML and AI
- Google BigQuery for tabular data storage
- Cloud BigTable for NoSQL database-style storage
- Connections to BI tools like Tableau, Looker and Power BI

Data visualisation and data analytics tools

Data visualisation tools allow you to view your data in a pictorial format (like graphs and charts), which makes it easier to draw coherent insights from it, thus simplifying the analytical process. Here are a few data visualisation and data analytics tools you can integrate into your business model (these include tools that are not open source).

 **Note:** Matplotlib, is an open source Python library which is also used for data visualising and plotting but requires a high level of coding knowledge. On the other hand, Power BI and Tableau are robust tools with user-friendly interfaces suitable for business intelligence and data visualisation.

Tableau: Tableau is an excellent data visualisation and business intelligence (BI) tool used for reporting and analysing vast volumes of data. It can easily connect to different data sources.

- It allows easy access to visualisations for partners, teams and clients.
- It allows you to create interactive maps automatically.
- It provides unlimited data exploration with interactive and intuitive dashboards.
- Dashboard setup hardly takes a few minutes with data from popular web applications.

Power BI: Microsoft Power BI provides interactive visualisations and BI (business intelligence) capabilities with a simple interface, designed to be used by analysts and data scientists alike.

- It offers a library of pre-built connectors.
- A simple drag-and-drop interface is provided by this tool.

The challenges

The need for massive data storage:

This is one of the most common issues, as enterprises need multiple platforms to store data flowing from multiple sources.

Duplication of data is inevitable due to multiple systems capturing the same information and storing it across multiple tables. Data deduplication techniques can help recognise identical points and flag them for duplication because different data source providers may write the same information in different ways.

Incomplete and incorrect data: If stakeholders are unable to access and use data in a productive way, then even the most advanced data management technologies are useless for a business.

If the data isn't displayed in an understandable dashboard that addresses pertinent queries and gives the necessary insights to the appropriate individuals, it won't be of any use.

If data is manually entered, fields may be incomplete or incorrect. The accuracy of data analysis is contingent upon the quality of the underlying data. This implies that human mistakes can occur in data.

Real-time data analysis

Data, in its raw form, has limited usefulness even if it is of high quality. Large-scale data analysis and extraction can benefit from technology, but there are still numerous obstacles to overcome, such as operating the tool appropriately and extracting the data logically, among other things.

With a real-time dashboard, you may access a multitude of sophisticated tools that offer insights into the functioning of your organisation. You require a platform that efficiently handles the unprocessed data and extracts insightful knowledge from it. 

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